

What we are gonna do:

RTL Design



Simulation



Synthesis



Floorplanning



Placement



Clock Tree Synthesis (CTS)



Routing



DRC/LVS Checks



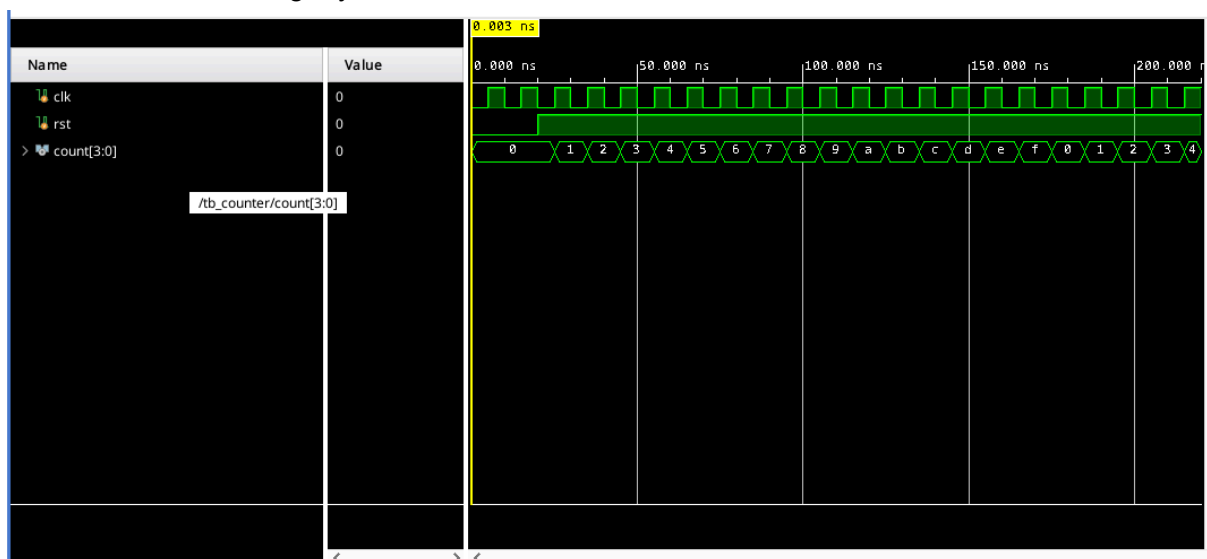
GDSII Layout

What I am doing now?

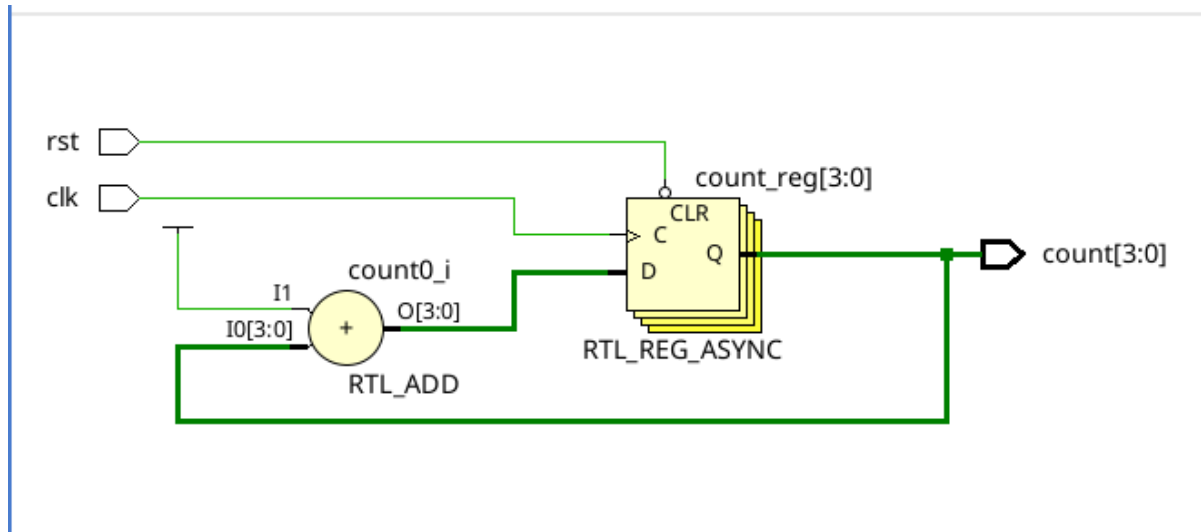
First I had created two directories , since i am working on rhel10 , I am saying them directories , one main directory and another its subsidiary directory where I will be storing my main verilog code.

In the linux terminal i have created a file counter.v in which I am designing it to be a 4 bit counter.Verilog is being provided in my github repo and it will be easier for you to see from there too. I had created testbench file for it also.

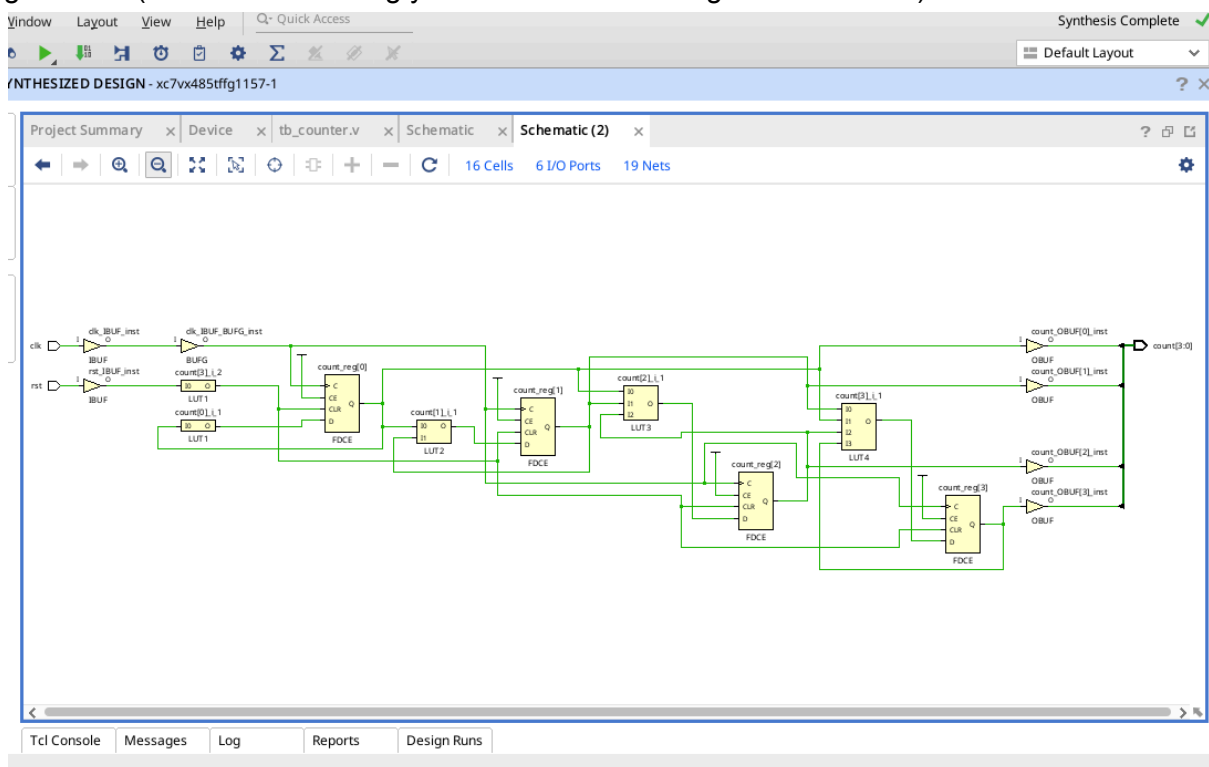
In vivado I will be doing my behavioral simulation and its result is shown below:



From the behavioural simulation we can say that our counter is working correctly. So let me generate its schematic diagram also shown that too to you:



After this I have run synthesis in my vivado suit let me provide details of that too what it generated(it will be converting your behavioral rtl into gate level netlist).



Now its time for me to shift to OpenLane.

Stage	Meaning
Synthesis	RTL → gates
Floorplan	chip dimensions
Placement	place cells
CTS	build clock network
Routing	connect wires
DRC	design rule checks
LVS	layout vs schematic
GDSII	final layout



What is PDK?

Process Design Kit(PDK) it is the manufacturing technology information provided by the foundries.

What PDK Contains

The PDK defines:

- transistor models
- metal layers
- design rules
- spacing rules
- standard cells
- parasitic models

Without a PDK:

you cannot manufacture silicon

OpenLane commonly uses SKY130 means : 130nm Skywater process Open source ASIC technology.

What I'll Do Next

We will:

1. create an OpenLane design directory
2. add my counter RTL
3. create config file
4. run RTL-to-GDS
5. inspect reports/layouts

So , in the OpenLane directory , I went straight to the design directory and in that i create a folder named counter where, I again created directory name src as openlane has its predefined file structure to process. And in src folder I have my rtl code.

What are "Standard Cells"?

A Standard Cell Library is a collection of pre-designed, blueprinted layout blocks provided by the semiconductor foundry (like TSMC, Intel, or Samsung).

Unlike the FPGA's fake memory-based logic, these are **actual, physical, hardwired transistors** optimized to be as tiny and fast as possible. The library includes:

- Basic gates (AND, OR, NAND, NOR, XOR)
- Storage elements (Flip-Flops, Latches)
- Complex cells (Multiplexers, Full Adders)

After that I created the config.json file in the counter directory.

VERY IMPORTANT ASIC CONCEPT

CLOCK_PERIOD

"CLOCK_PERIOD": 10.0

means:

design must run at 100 MHz

because:

Frequency = 1 / Period

So:

$1 / 10\text{ns} = 100\text{ MHz}$

Why Timing Matters

Backend tools must ensure:

signals settle before next clock edge

This is:

timing closure

Now after making config file you will be running the OpenLane , by going to the OpenLane directory and then use command make mount. This will open up openlane shell. Inside it write

```
./flow.tcl -design counter
```

UNDERSTAND EACH STAGE

1. Synthesis

RTL → standard cells

using:

- NAND gates
- flip-flops
- buffers

2. Floorplanning

Defines:

- chip dimensions
- IO locations
- power grid

3. Placement

Places cells physically on silicon.

4. CTS

Clock Tree Synthesis.

Builds clock routing network.

VERY important backend topic.

5. Routing

6. DRC

Checks manufacturing rules.

7. LVS

Checks:

layout == schematic

8. GDSII

Final chip layout file.

This is:

actual manufacturable geometry

Tap/Decap Insertion

This inserts:

- substrate taps
- decoupling capacitors

Important for:

- power integrity
- latchup prevention
- noise stability

